

QUESTION BANK BASED ON PYQ

❖ Long Questions (5 to 8 marks)

1. Discuss LAN and WAN.
2. Write a note on Satellite Communication.
3. Write a note on Wireless Network.
4. Explain network types based on Geographical network.
5. Define Protocol. Explain different types of network protocol.
6. Write a note on OSI Reference Model.
7. Write a short note on TCP and UDP.
8. Define Segment? Explain each field of TCP segments.
9. Write a note on connection less protocol with its applications.
10. Explain addressing scheme at the Network Layer of IP.
11. Explain User Datagram with its applications.
12. Write a note on TCP with its header.
13. Write a note on TCP Operation.
14. Write a note on UDP.
15. Explain different topologies with its advantages and disadvantages.
16. Discuss need, uses and advantages of Network.
17. Discuss peer network with its advantages and disadvantages.
18. Discuss Bus Topology in detail.
19. Explain Star topology in detail.
20. Explain Star and Ring Topology in detail.
21. Write a short note on ring topology.
22. Write a note on Mesh topology.
23. Differentiate between IP address and MAC address.
24. Explain the responsibility of three Upper layers of OSI model.
25. Explain each layers of OSI reference model.
26. Define unguided media. Discuss each class of unguided media.
27. Write a note on Coaxial cable.
28. Explain Optical Fibre in brief.
29. Write a note on Fiber optics cable with its advantages and disadvantages.
30. Explain network and transport layer of OSI model.
31. Discuss different protocol of transport layer.
32. Discuss in detail about layers of OSI model.
33. Explain wireless transmission media.
34. Write a note on twisted pair cable.
35. How Switches differ from Hub?
36. What is Header? Explain each field of TCP header.
37. Explain Application Layer and Presentation Layer components and functions.
38. Mention the advantages and disadvantages of Fibre optics.
39. Explain in detail how the information flows from one host to another in OSI based system.
40. Explain the responsibilities of application and presentation layer of OSI reference model.

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41. Explain Transmission Media. Explain in detail Guided Media with its application, advantages and disadvantages. (8 marks-2016)
42. Write a note on IP address and IP Subnet. (8 marks-2016)
43. Explain Internet Transport Protocols in detail. Explain TCP with its features and services. (8 marks-2016)
44. What is an IP address? Explain different classes of IP address. Also explain the default mask of each category of IP classes. (8 marks-2017)
45. What is Network Hub? Write its applications. Explain the different types of Hub used for network with diagram. (8 marks-2017)
46. Explain the Bus Topology and Ring type topology networks. Compare their performances. (8 marks-2017)
47. What is Firewall? What are the functions of Firewall? Explain the type of firewall in detail. (8 marks-2017)
48. Write a short note on TCP/IP model.
49. Write a short note on Bridges.
50. Write a Short note on Wireless Network.
51. Explain MANET and its types of network.
52. Where FANET is used and how?
53. How search engines work?
54. What is URL?
55. Define URL and write steps to access URL.
56. Write a detailed note on Protocol used for accessing URL.
57. Explain working of mail server.
58. Explain working of email account.
59. Explain POP and SMTP Protocol.
60. What is Domain Name Server?

❖ Short questions (2 or 4 marks)

1. Define analog and digital signals.
2. Write advantages and disadvantage of Bus topology.
3. Define Network Protocols.
4. Define the term Computer Network.
5. What are the advantages of Star topology?
6. Explain different use of Network.
7. Define Bandwidth and Transmission rate?
8. Explain BitRate and Nyquist bit rate.
9. Differentiate between Star and Ring topology.
10. What is the function of SMTP and HTTP?
11. What do you mean by encryption and decryption?
12. Explain the term: Frequency and Nyquist Bit Rate.
13. Write the differences between Baseband and Broadband transmission.
14. Write the differences between Repeater and Hub.
15. Explain Token Passing Mechanism in Ring Topology.
16. Explain Guided Communication Media. List any three Examples.
17. Differentiate Guided and Unguided media.
18. Why there is the twist in the twisted pair cable? Justify. **OR**
19. Why do we require a twist in a twisted pair cable?

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20. Define term: Baud Rate and Bandwidth.
21. Define client and server.
22. What is Network?
23. Define WAN.